Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Robust Clinical Information Extraction from Free-Text Medical Notes for Downstream Decision Support

PI name *

Mamatha Bhat

Medical field

Liver Transplant

Description (paragraph) *

From the clinical notes, students could extract structured variables in stages: starting with general features such as demographics, then adding comorbidities, and finally incorporating liver-specific variables.
Tasks:
Define a schema of variables (demographics, comorbidities, relative-contraindication to transplant, and other liver-specific factors).
Define extraction targets and temporal scope.
Compare baseline methods (regex / dictionaries / ontologies like UMLS / SNOMED) with supervised (ClinicalBERT) and LLM based (GPT) extraction methods
Evaluation using standard metrics such as Precision, recall, F1-score, and new metrics for error taxonomy (missingness, hallucination, temporal misalignment)
Advance task: Low-resource learning (small labeled set)

Desired outcome *

Deliver a reusable extraction pipeline for Liver transplant patients

Data Type (click all that apply)

- [x] Tabular (clinical, EHR, registries)
- [ ] Imaging (MRI, CT, X-ray, histopathology)
- [ ] Time series / signals
- [ ] Genomics / omics
- [ ] Text / clinical notes
- [ ] Simulation / synthetic data
- [ ] Other:

Data Availability & Access *

- ( ) Publicly available
- (•) Requires ethics approval
- ( ) Available within my group
- ( ) Simulated / generated
- ( ) Other:

Relevant Skills / Background of the Student Team

Machine learning experience, familiarity with NLP

Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Machine Learning Medical Directive

PI name *

Anna Goldenberg

Medical field

Emergency Department

Description (paragraph) *

A Machine Learning Medical Directive is a system that uses predictive models to recommend medical tests or imaging for patients as soon as they complete the triage phase in the emergency department. Instead of waiting for a doctor's approval, it allows physician assistants to act quickly under established medical directive rules, reducing delays in care. By analyzing patient information like symptoms and vital signs, the system suggests likely needed tests, such as X-rays or ultrasounds, to speed up diagnosis. This approach improves efficiency, shortens wait times, and supports clinicians without replacing their judgment. Ultimately, it helps deliver faster, safer, and more effective care for patients.

Desired outcome *

1- Improved Labeling strategy using Language models, 2- Design a new models based on language models and compare the results to the existing baseline

Data Type (click all that apply)

- [x] Tabular (clinical, EHR, registries)
- [ ] Imaging (MRI, CT, X-ray, histopathology)
- [ ] Time series / signals
- [ ] Genomics / omics
- [x] Text / clinical notes
- [ ] Simulation / synthetic data
- [ ] Other:

Data Availability & Access *

- ( ) Publicly available
- ( ) Requires ethics approval
- (•) Available within my group
- ( ) Simulated / generated
- ( ) Other:

Relevant Skills / Background of the Student Team

Machine Learning, Natural language processing, Master's level Critical thinking

Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Benchmarking Multi-Modal Fusion Strategies for Relapse Prediction in Bipolar Disorder using Wearables and Sentiment Data.

PI name *

Dr. Venkat Bhat

Medical field

Psychiatry / Affective Disorders.

Description (paragraph) *

Predicting clinical relapses (manic or depressive episodes) remains a major challenge. In 2026, the proliferation of wearable sensors and social sentiment analysis offers a unique opportunity for early intervention. This project will evaluate transformer-based fusion techniques that combine physiological signals (heart rate, sleep patterns) with linguistic sentiment from digital journals or clinical notes to identify subtle "digital biomarkers" of impending relapse.

Desired outcome *

A comparative technical paper or conference submission evaluating the performance gains of multi-modal fusion over single-modality baselines.

Data Type (click all that apply)

- [ ] Tabular (clinical, EHR, registries)
- [ ] Imaging (MRI, CT, X-ray, histopathology)
- [x] Time series / signals
- [ ] Genomics / omics
- [x] Text / clinical notes
- [ ] Simulation / synthetic data
- [ ] Other: ____________________

Data Availability & Access *

- (●) Publicly available
- ( ) Requires ethics approval
- ( ) Available within my group
- ( ) Simulated / generated
- ( ) Other: ____________________

Relevant Skills / Background of the Student Team

Deep learning, time-series analysis, and basic Natural Language Processing (NLP)

Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Investigating Demographic Bias and Algorithmic Fairness in Automated Depression Screening Tools for Diverse Populations

PI name *

Dr. Venkat Bhat

Medical field

Public Health / Mental Health Screening

Description (paragraph) *

As of 2026, global regulations are tightening around AI-driven clinical consultations to prevent harm and ensure equity. This project will audit a state-of-the-art diagnostic model (e.g., screening for depression via speech or text) to identify performance disparities across different demographic groups (age, gender, ethnicity). The student will implement and test "fairness-aware" machine learning techniques to mitigate identified biases.

Desired outcome *

A research paper detailing a bias audit and proposing a mitigation framework

Data Type (click all that apply)

- [ ] Tabular (clinical, EHR, registries)
- [ ] Imaging (MRI, CT, X-ray, histopathology)
- [ ] Time series / signals
- [ ] Genomics / omics
- [x] Text / clinical notes
- [ ] Simulation / synthetic data
- [x] Other: Audio (Speech signals)

Data Availability & Access *

- (•) Publicly available
- ( ) Requires ethics approval
- ( ) Available within my group
- ( ) Simulated / generated
- ( ) Other:

Relevant Skills / Background of the Student Team

Statistical analysis of fairness metrics, Python, and familiarity with AI ethics principles

Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Evaluating the Accuracy, Reliability, and Safety of Large Language Models for Detecting Suicidal Ideation and Self-Harm Signals in Text Conversations

PI name *

Dr. Venkat Bhat

Medical field

Mental health screening; suicide prevention; digital health safety evaluation

Description (paragraph) *

Increasingly, individuals use general-purpose LLMs for health and mental health questions. A key safety concern is whether an LLM (or an LLM-based "screener") can reliably detect text-based signals of suicidal ideation and self-harm and trigger appropriate escalation at the right time. This project evaluates multiple routinely available LLMs as screeners, with emphasis on sensitivity, calibration, robustness, and safe escalation behavior.

Desired outcome *

A research paper reporting comparative performance of LLMs for suicidality/self-harm signal detection and escalation, including error analyses and recommendations for safe deployment constraints

Data Type (click all that apply)

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- [ ] Imaging (MRI, CT, X-ray, histopathology)
- [ ] Time series / signals
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- [ ] Text / clinical notes
- [x] Simulation / synthetic data
- [x] Other: • Public, de-identified conversation datasets with counseling-style dialogues

Data Availability & Access *

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- ( ) Other:

Relevant Skills / Background of the Student Team

Experimental design, statistical evaluation (classification + time-to-event), Python, prompt engineering, robust benchmarking, and multi-agent orchestration.

Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Estimating Abnormal Central Venous Pressure Utilizing Photoplethysmography

PI name *

Asad Siddiqui

Medical field

Anesthesia/Critical Care

Description (paragraph) *

Central Venous Pressure (CVP) is an important marker of health and volume status in critically ill patients. The measurement of CVP requires invasive access that is technically challenging and has associated morbidities and can only be measured in high intensity units such as the operating room or intensive care unit. Photoplethysmography (PPG) is a non-invasive tool that is readily available in hospital and non-hospital settings. PPG waveforms have previously been used to identify patients that would be responsive to receiving volume. Recent literature has demonstrated that PPG waveforms can be used to estimate abnormal CVP readings in adults (doi: 10.64898/2025.12.30.25343231.). At the Hospital for Sick Children, we have a large repository of physiologic waveform data available from our critical care unit and operating room. The goal of this project would be to train a model to estimate abnormal (low and high) CVP values utilizing non-invasive waveform signals (PPG).

Desired outcome *

Develop a deep learning model to estimate abnormal CVP values utilizing PPG waveform signals

Data Type (click all that apply)

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- [ ] Other:

Data Availability & Access *

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- ◉ Available within my group
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- ○ Other:

Relevant Skills / Background of the Student Team

Working with time-series data

Proposed Project for LMP2392H

Please let us know your suggested project topics for students to choose from.

Project title *

Predicting ICU Morbidity Outcomes in Liver Transplant Patients Utilizing Physiologic Data

PI name *

Asad Siddiqui

Medical field

Anesthesia/Critical Care

Description (paragraph) *

Pediatric liver transplantation offers excellent long-term survival for many pediatric liver diseases; however, the perioperative period is associated with the highest morbidity and mortality, driven by the multisystem effects of liver disease and the physiologic stress of transplantation. These high-risk patients require intensive perioperative monitoring, including continuous arterial and central venous pressure waveforms, to support expert bedside interpretation and early identification of threats to the transplanted graft and other vital organs. Advances in monitoring technology now enable acquisition of high-frequency physiologic data (>1 kHz), creating opportunities for real-time assessment of patient status and the development of machine learning–based decision support systems for dynamic risk prediction. Prior work has demonstrated that continuous physiologic variables, including nontraditional metrics such as heart rate variability, are associated with organ dysfunction, mortality, and ICU length of stay in high-risk populations. In pediatric liver transplant recipients, prolonged ICU length of stay—particularly beyond 7 days—is associated with increased morbidity, including longer ventilation duration, higher rates of return to the operating room, graft rejection, and mortality, which remains approximately 4.9% at 90 days. The goal of this project will be to utilize a database of liver transplant patients (approx. 250) and high-frequency physiologic waveform data stored collected from the critical care to determine if early ICU physiologic data (immediately post-operative 6 hours) can be used to enhance risk scores to predict ICU LOS.

Desired outcome *

Determine the impact of applying continuous physiologic data to static risk scores to predict morbidity (intensive care unit length of stay and composite morbidity outcome) in liver transplant patients

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Data Availability & Access *

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Relevant Skills / Background of the Student Team

Ability to use clinical multimodal data (EMR data and time-series waveform data)